Chapter 7

Trees

Trees

Let A be a set, and let T be a relation on A.

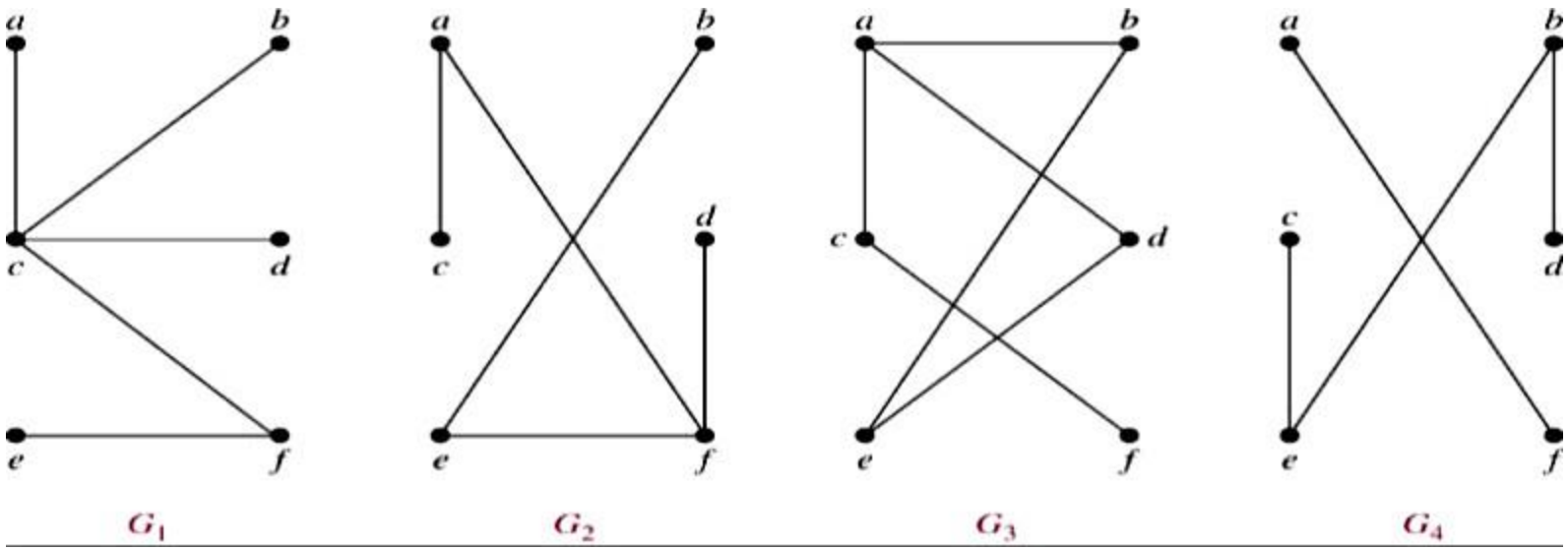
We say that T is a tree if there is a vertex x in A with the property that

- There is **a unique path** in T from x to every other vertex in A.
- No path from x to x .

Usually, x is called the root of the tree, and T is referred to as a rooted tree (denoted by (T, x))

Def. A tree is a <u>connected</u> undirected graph with <u>no</u> simple <u>circuits</u>.

Example Which of the graphs are trees?



Sol: G_1, G_2

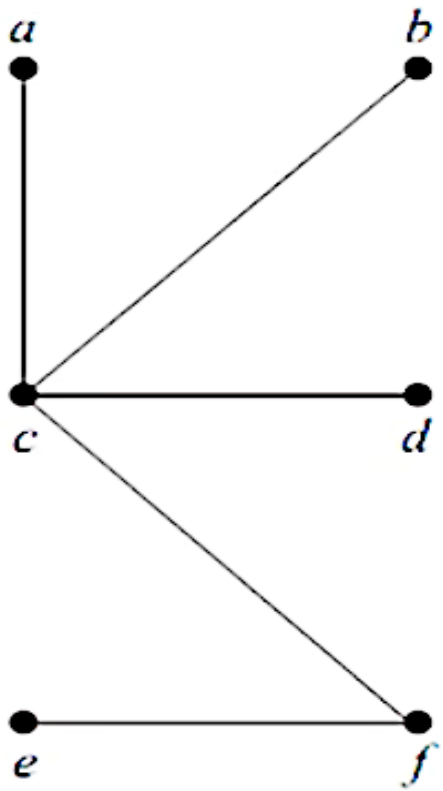
Note. If connected condition is removed, it becomes forest

Example 1

Example: Determine if the following graph is a tree?

- **Connected : Yes**
- **Has Circuit : No**

$\therefore$ G_1 is a tree.



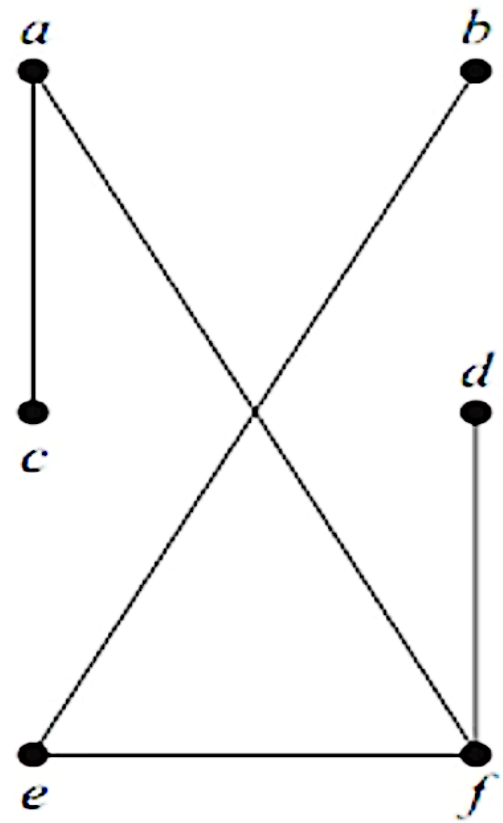
G_1

Example 2

Example: Determine if the following graph is a tree?

- **Connected : Yes**
- **Has Circuit : No**

$\therefore$ G_2 is a tree.



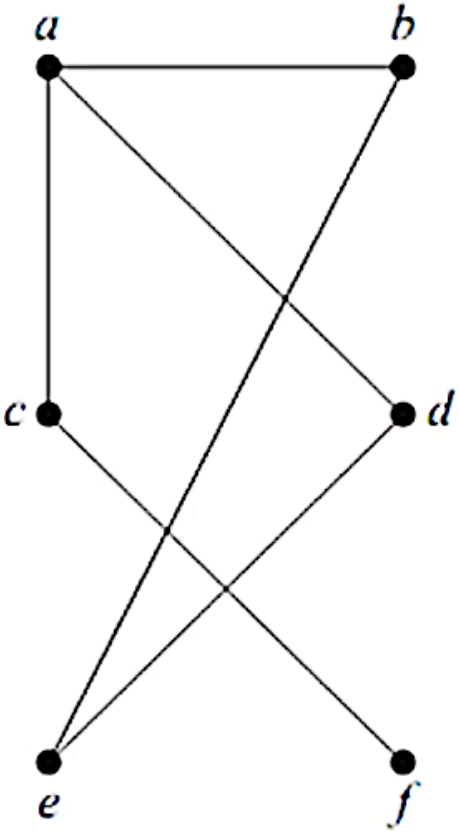
G_2

Example 3

Example: Determine if the following graph is a tree?

- **Connected : Yes**
- **Has Circuit : Yes**

∴ G_3 is NOT a tree.



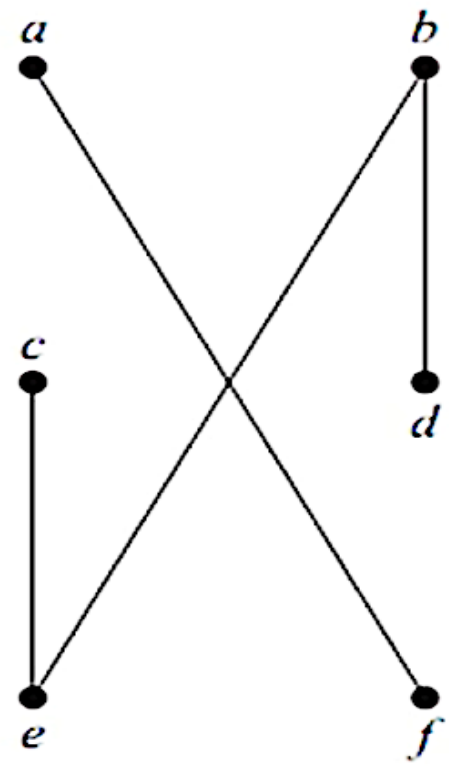
G_3

Example 4

Example: Determine if the following graph is a tree?

- **Connected : No**

∴ G_4 is NOT a tree.



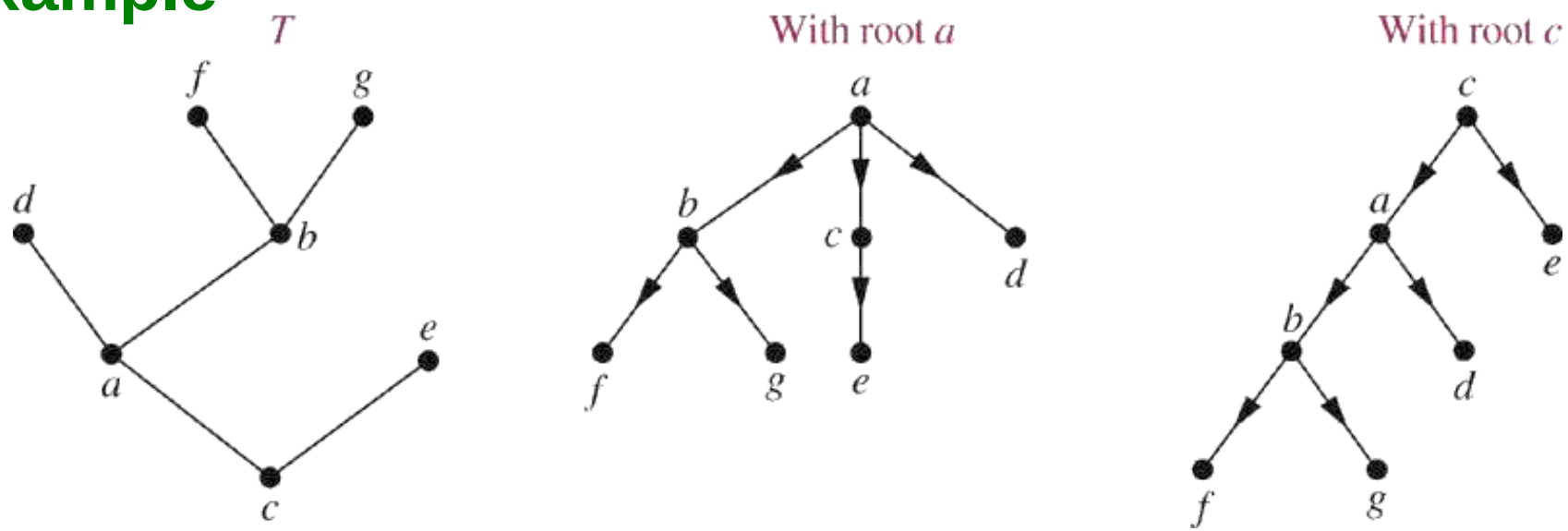
G_4

It is a **forest** if the graph has more than one tree

Any undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

Def 2. A **rooted tree** is a tree in which one vertex has been designed as the root and every edge is directed away from the root. (Arrow to disappear)

Example



Trees

- Theorem 1

  Let (T, x) be a **rooted tree**. Then

  (a) There are no cycle in T

  (b) x is the only root of T

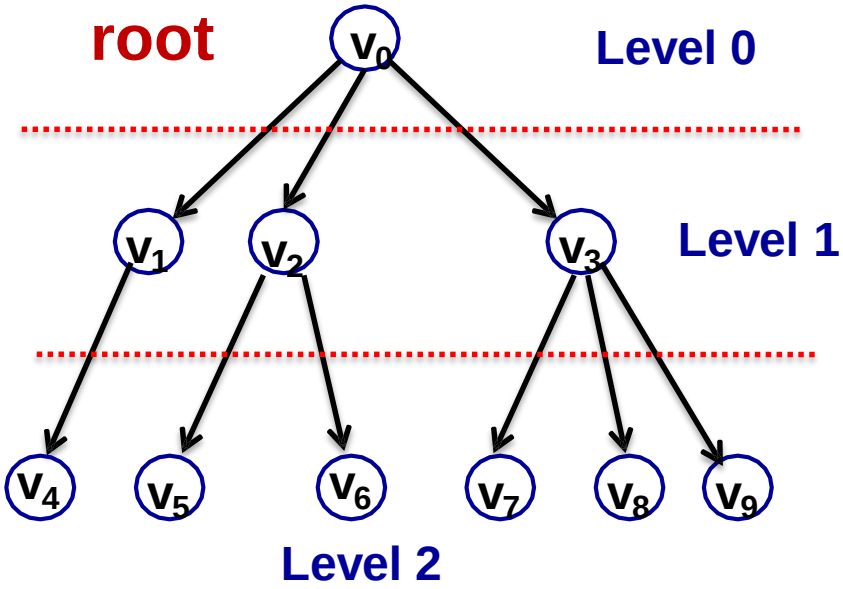
  (c) Each vertex in T, other than x, has in-degree one, and x has in-degree zero.

Root has zero parent

Each vertex rather than root has one parent

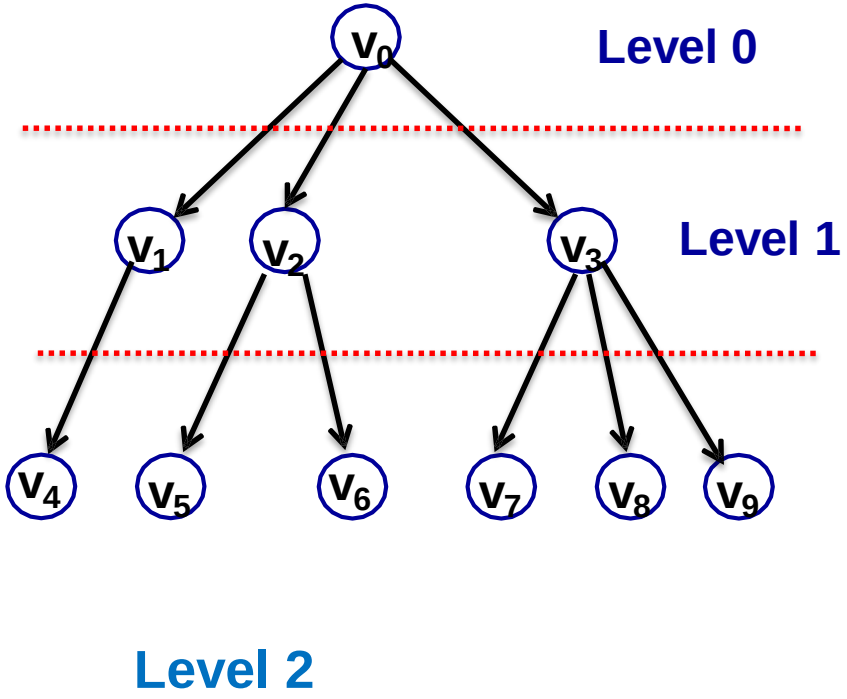
Trees

- **Level 0 vertices:** the vertices at level 0.
- **Level k vertices:** the vertices of k level.
- Example: level 2 vertices: v4, v5, v6, v7, v8, v9
- **Height of a tree:** the largest level number of a tree. Here, height=2.



Relation as a tree

R = {(v0,v1),(v0,v2),(v0,v3),
(v1,v4),(v2,v5),(v2,v6),
(v3,v7),(v3,v8),(v3,v9)}



Trees

- **Parent-offspring**:

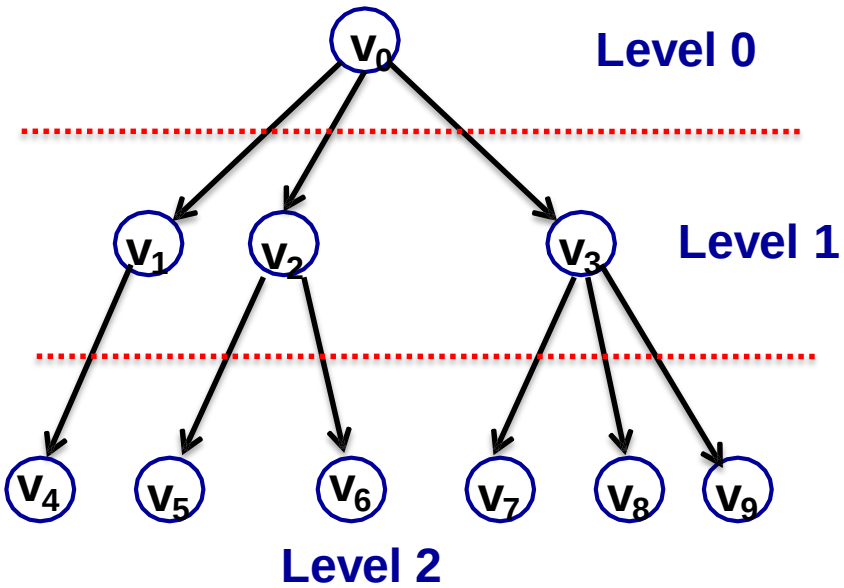
For all pairs (a, b) in T, a is called the parent of b, and b is called the offspring of a.

- **Siblings**:

The vertices that have the same parent

- **Leaves**:

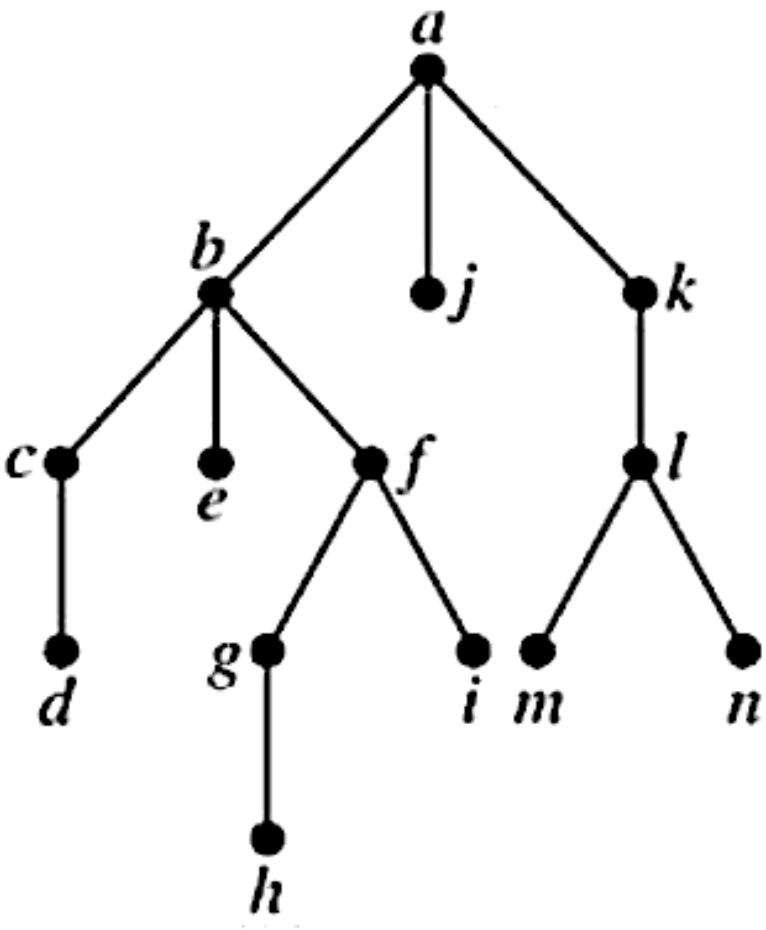
The vertices that have no children.



- **internal vertices:**

Vertices that have children.

Example



level

0

1

2

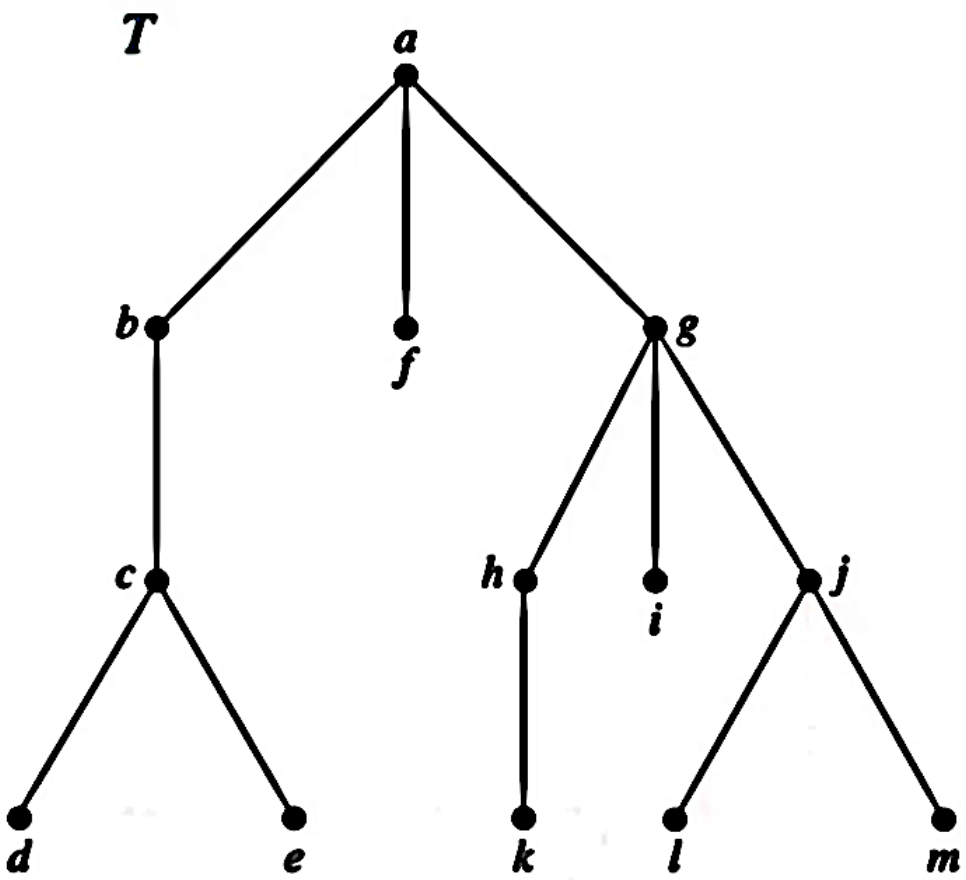
3

4

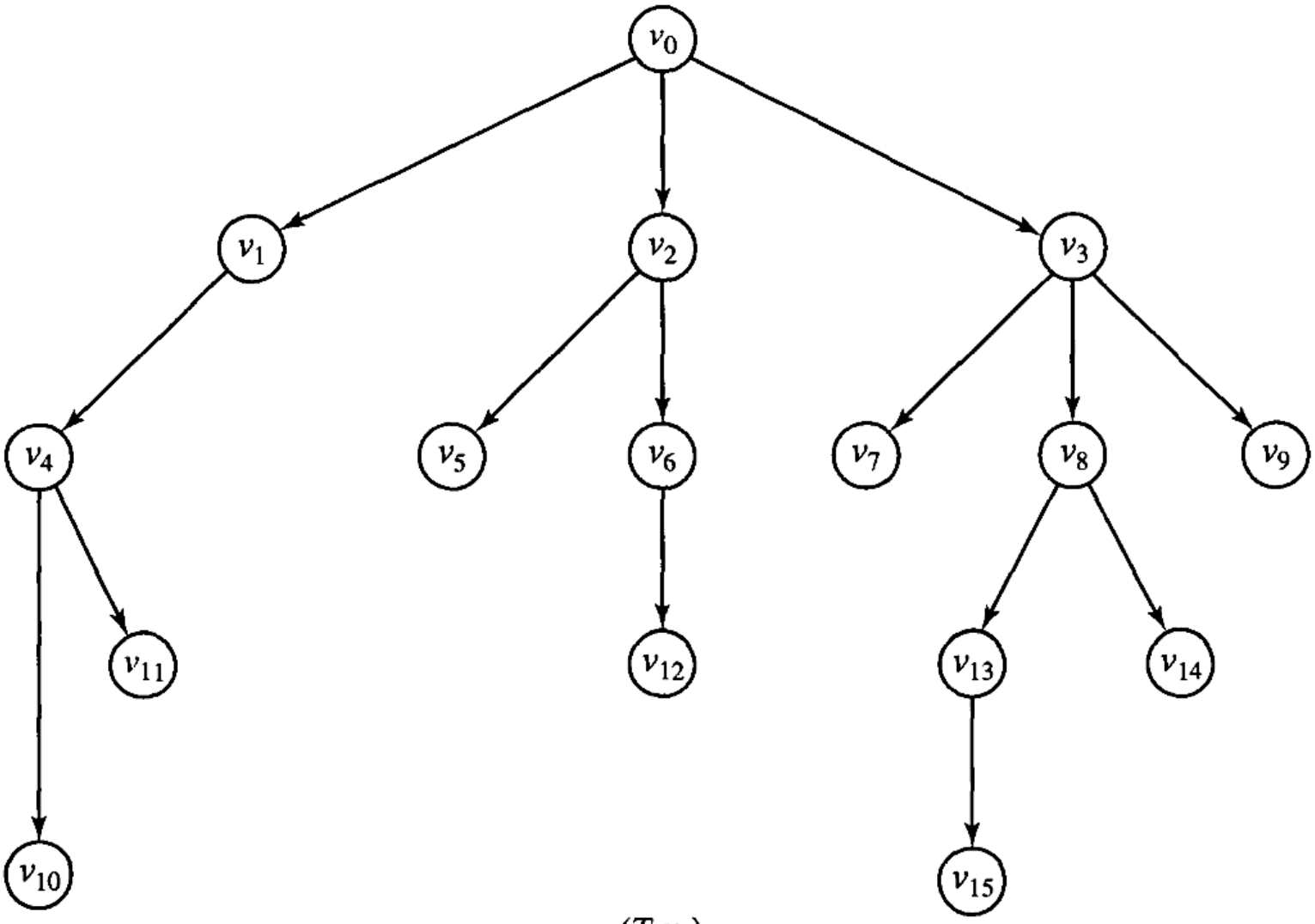
height = 4

Rooted tree

- Find parent of c.
  {b}
- Find children of g.
  {h, i,j}
- Find siblings of h.
  {i,j}
- Find ancestors of e.
  {c, b,a}
- Find descendants of b.
  {c,d,e}
- Find all internal nodes.
  {a,b,g,c,h,j}
- Find all leaves (external nodes).
  {f,d,e,I,k,l,m}



Question



(T, v_0)

Answer the following

List all level-3 vertices?

v10, v11, v12, v13, v14

List all leaves?

v10, v11, v5, v12, v7, v15, v14, v9

What are siblings of v8?

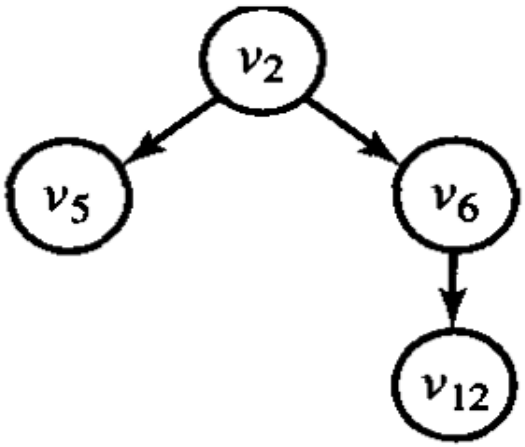
v7, v9

What are descendants of v8?

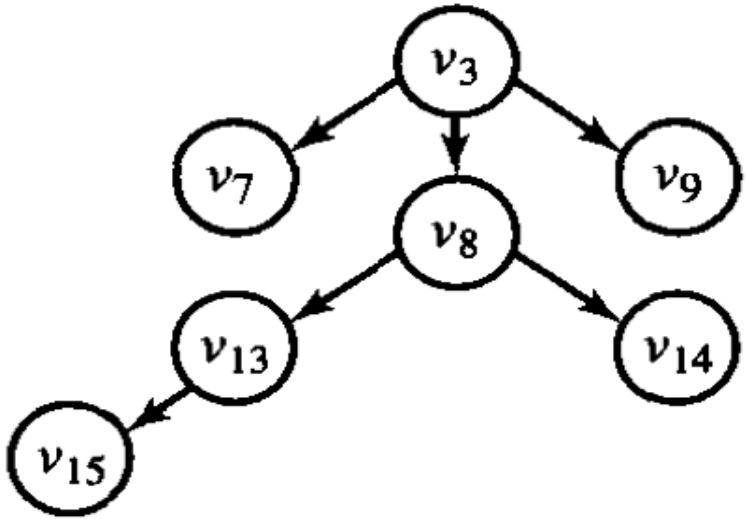
v13, v14, v15

(a) Compute the tree $T(v_2)$.
(b) Compute the tree $T(v_3)$.

(a)



(b)



(a) What is the height of (T, v_0)? a) 4

(b) What is the height of $T(v_3)$? b) 3

Is (T, v_0) an n-tree? If so, for what integer n?
Is (T, v_0) a complete n-tree? If so, for what integer n?

Sol.

(T, v_0) may be an n-tree for $n \geq 3$. It is not a complete 3-tree.

Theorem

Let (T, *x*) be a rooted tree on a set A. Then

(a) T is irreflexive

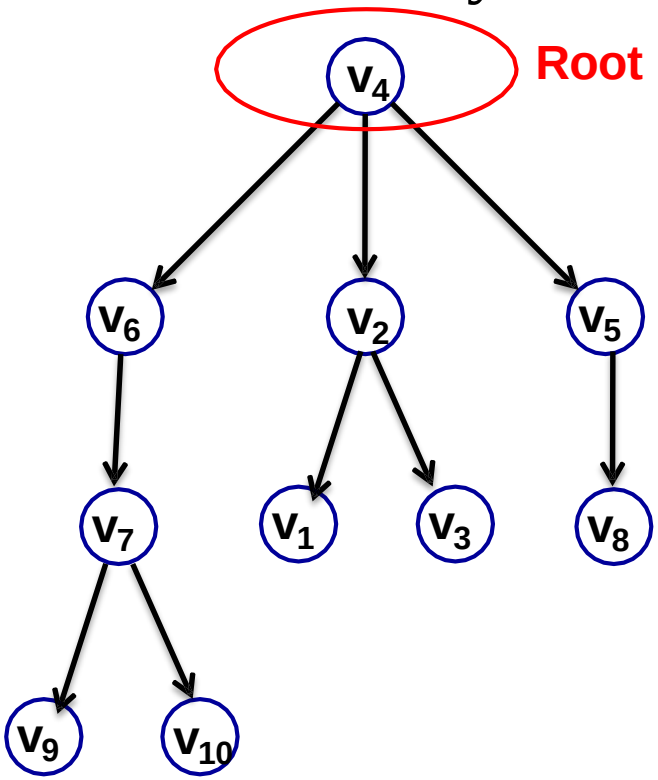
(b) T is asymmetric

(c) If (a,b) in T and (b,c) in T, then (a,c) is not in T, for all a, b and c in A. → not transitive

Trees

- Example

Let $A=\{v_1,v_2,\ldots v_{10}\}$ and let $T=\{(v_2,v_3), (v_2,v_1), (v_4,v_5), (v_4,v_6), (v_5,v_8), (v_6,v_7), (v_4,v_2), (v_7,v_9), (v_7,v_{10})\}$. Show that T is a rooted tree and identify the root.



In Exercises 1 through 8, each relation R is defined on the set A. In each case determine if R is a tree and, if it is, find the root.

1. $A = \{a, b, c, d, e\}$
$R = \{(a, d), (b, c), (c, a), (d, e)\}$

2. $A = \{a, b, c, d, e\}$
$R = \{(a, b), (b, e), (c, d), (d, b), (c, a)\}$

3. $A = \{a, b, c, d, e, f\}$
$R = \{(a, b), (c, e), (f, a), (f, c), (f, d)\}$

4. $A = \{1, 2, 3, 4, 5, 6\}$
$R = \{(2, 1), (3, 4), (5, 2), (6, 5), (6, 3)\}$

5. $A = \{1, 2, 3, 4, 5, 6\}$
$R = \{(1, 1), (2, 1), (2, 3), (3, 4), (4, 5), (4, 6)\}$

6. $A = \{1, 2, 3, 4, 5, 6\}$
$R = \{(1, 2), (1, 3), (4, 5), (4, 6)\}$

1. Yes, the root is b.

3. Yes, the root is f.

5. No.

Irreflexive, asymmetric, not transitive as child can't be a parent of its parent.

Trees

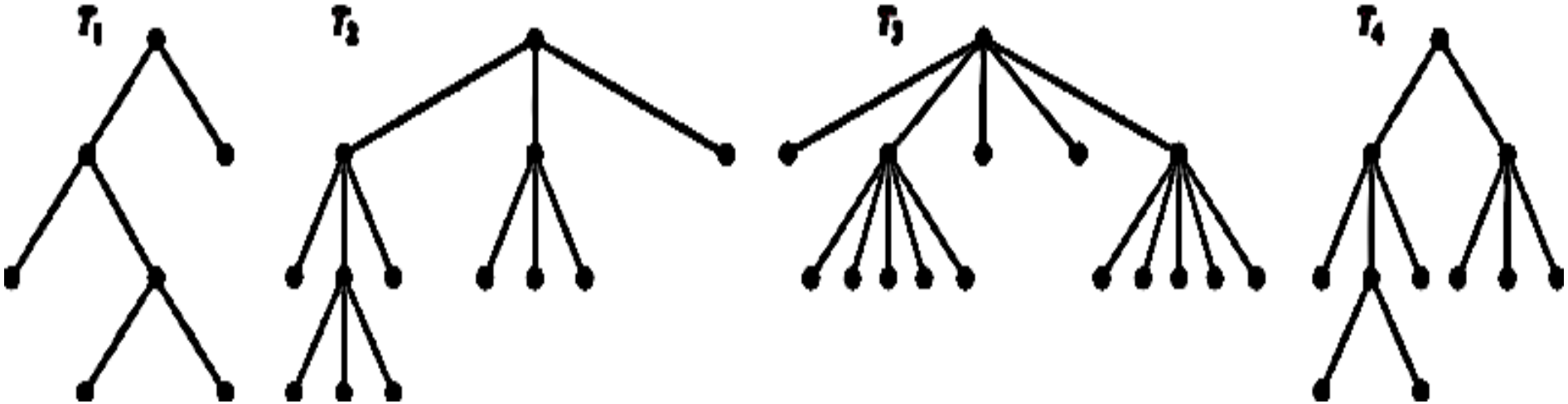
- **n-tree**

  If n is a positive integer, we say that a tree is an n-tree if every vertex **has at most n offspring**. In particular, a 2-tree is called a binary tree.

- **Complete n-tree**

  If all vertices of T, other than the leaves, have exactly n offspring, we say that T is a complete n-tree. A complete 2-tree is called a completed binary tree.

Example



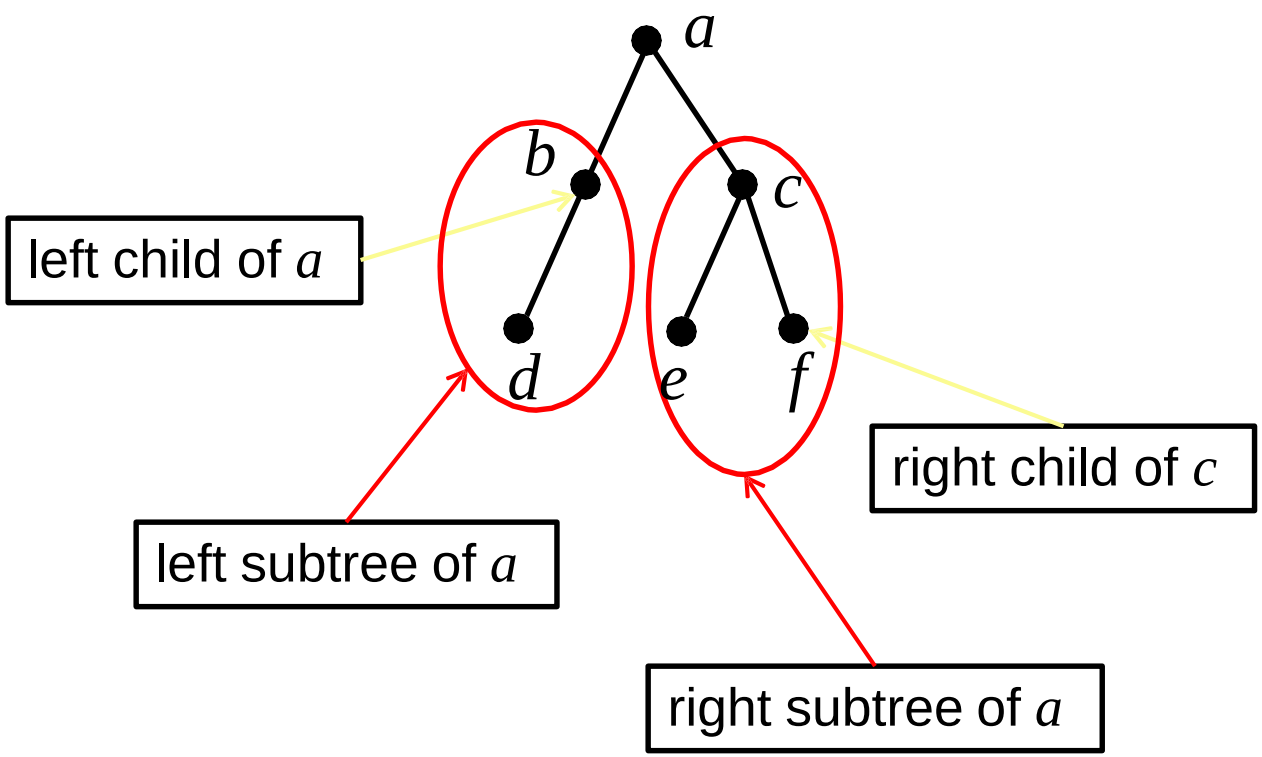
Complete binary tree

Complete 3- tree

Complete 5- tree

not complete 3-tree

Def:


a
b
c
d
e
f
left child of a
right child of c
left subtree of a
right subtree of a

Properties of Trees

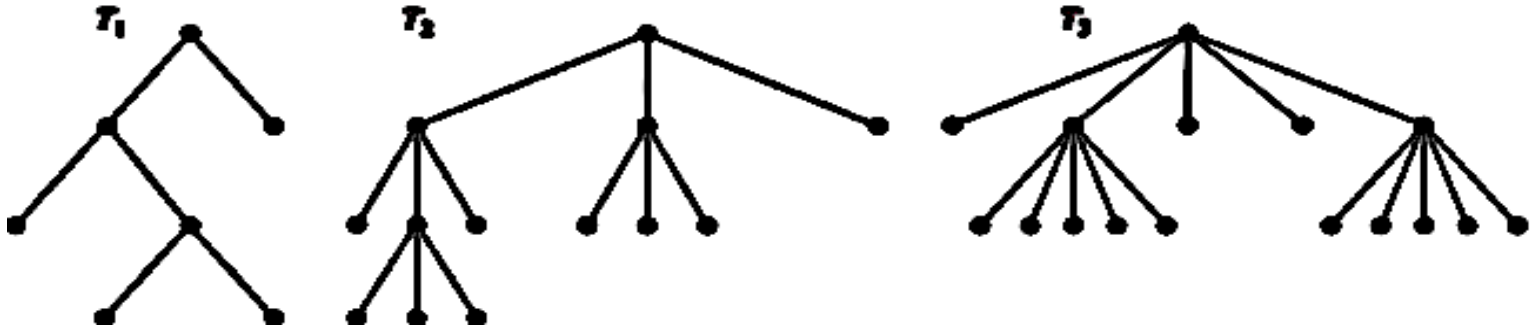
A tree with n vertices has $n-1$ edges.

A complete m-tree with i internal vertices contains $n = mi + 1$ vertices.

A complete m-tree with n vertices contains $(n-1)/m$ internal vertices, and hence $((m-1)n+1)/m$ leaves.

Example

A complete m-tree with i internal vertices contains $n = mi + 1$ vertices.



Complete binary tree

n=2*3+1=7

Complete 3- tree

n=3*4+1=13

Complete 5- tree

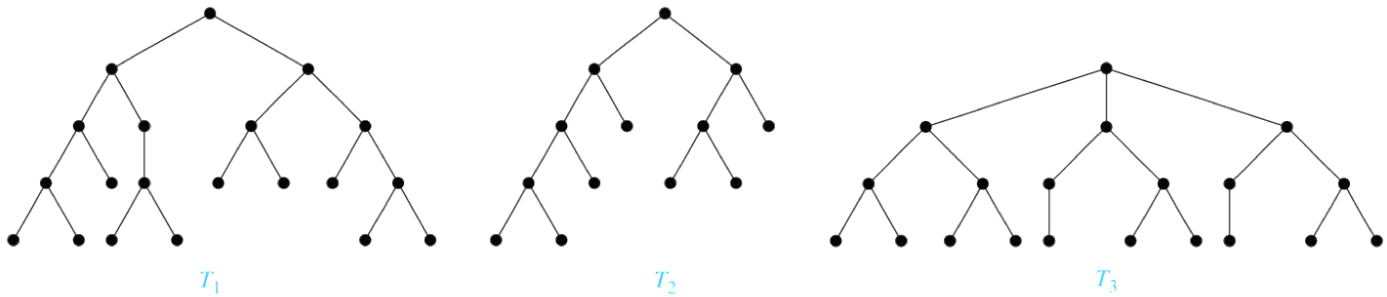
n=5*3+1=16

- A a complete 3-tree with 4 internal vertices has a number of vertices equal to 13.
- A a complete 5-tree with 3 internal vertices has a number of vertices equal to 16.

Balanced tree

A rooted tree of height *h* is *balanced* if all leaves are at levels *h* or *h* − 1.

Example: Which of the rooted trees shown below is balanced?



Solution: T_1 and T_3 are balanced, but T_2 is not because it has leaves at levels 2, 3, and 4.

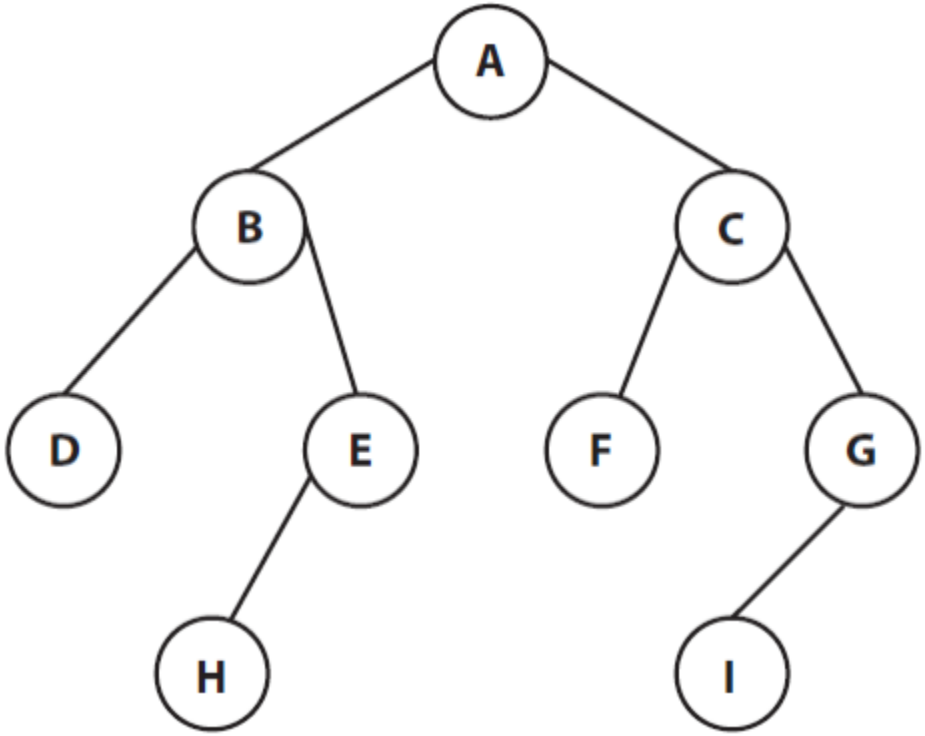
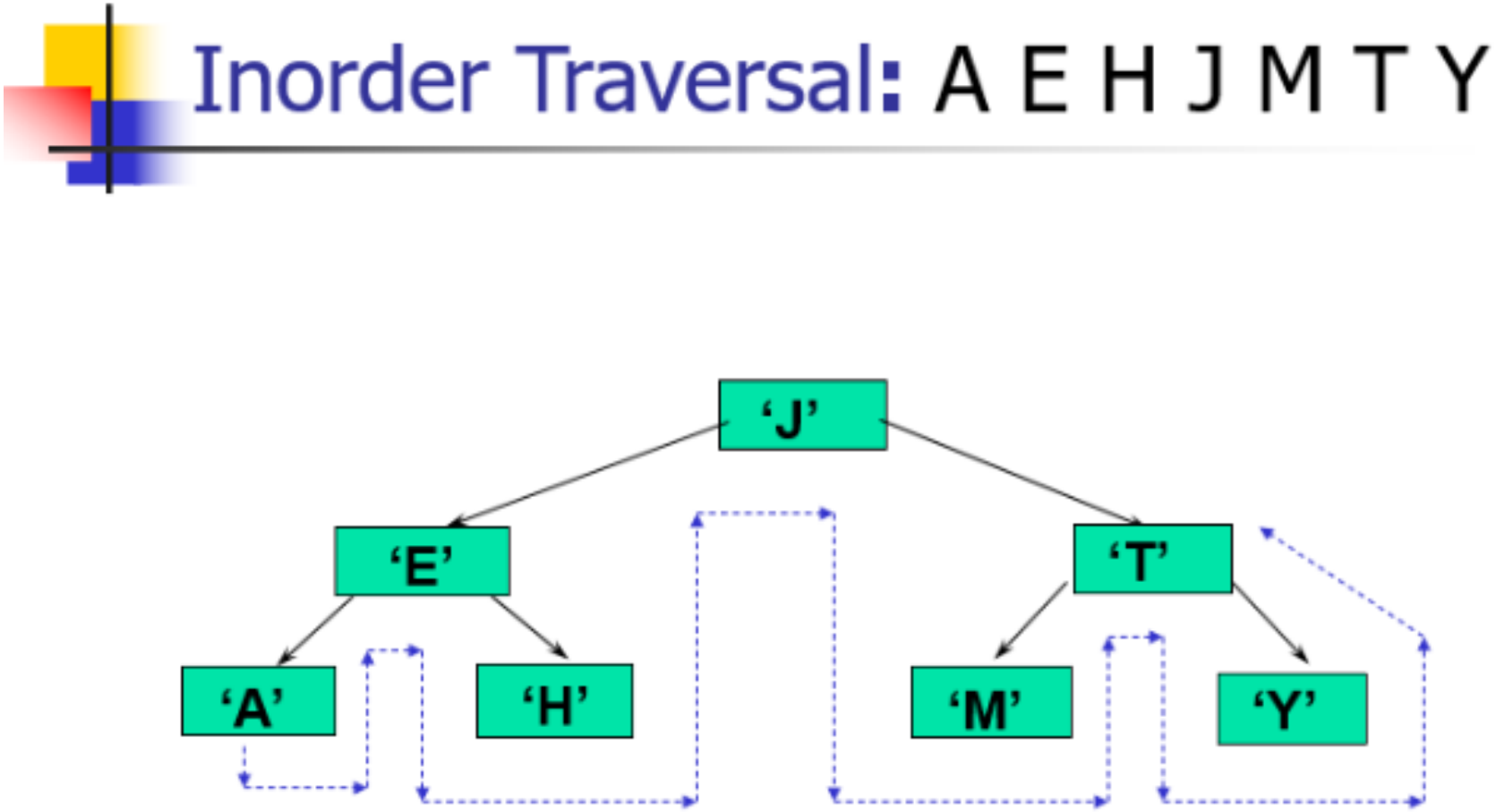
Traversing with Tree

- The tree traversing is the way to visit all the nodes of the tree on a specific order.
- The Tree traversal can be accomplished in four different ways:
  - Inorder Traversal
  - Pre Order Traversal.
  - Post Order Traversal.

Inorder Traversal

- Traverse the Left Subtree in INORDER(Left)
- Visit the Root node
- Traverse the Right Subtree in INORDER(Right)

Inorder : **D B H E A F C I G**



Inorder Traversal